

TOSM-VLA: Semantic World Modeling for Role Assignment and Recovery in Neurosymbolic Bimanual Manipulation

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Abstract—Bimanual manipulation remains a challenging problem in robotics, requiring precise coordination between two arms to execute long-horizon tasks. We present TOSM-VLA, a neurosymbolic framework that integrates a Triplet Ontological Semantic Model (TOSM) with a Vision-Language-Action (VLA) policy for bimanual robotic manipulation. Our approach employs a closed-loop architecture in which a TOSM-based world model persistently governs execution, generating semantic task plans, constraining arm role assignment and grasp selection within physical and semantic limits, monitoring execution state against relational TOSM signatures, and triggering cause-conditioned recovery upon failure detection, enabling the robot to adapt to unexpected perturbations throughout long-horizon bimanual tasks without task restart. We evaluate our system on a dual SO-101 robot platform across desk organization tasks requiring coordinated two-arm manipulation. Preliminary experiments on a desk organization task achieve 90% semantic task planning accuracy, 82% failure detection rate, and 64% recovery success rate, demonstrating the viability of integrating deterministic world modeling with reactive VLA policies for robust long-horizon bimanual manipulation.

I. INTRODUCTION

For decades, robotic manipulation required the world to conform to the robot, with fixed environments, known geometries, and hand-engineered plans. The robot knew where objects were, never what they were. Bimanual manipulation exposed this most sharply: which arm approaches and which supports were encoded by the human demonstrator, never derived from object semantics. RT-2 [1] first demonstrated that VLA models could transfer web-scale semantic knowledge to robotic control, and subsequent work including Octo [2], OpenVLA [3], and π_0 [4] established that robots could follow language instructions, reason about novel objects, and generalize across tasks. Yet two fundamental limitations persist. VLAs lack a persistent semantic world model—each observation is processed in isolation, with no structured

memory of object properties, spatial relationships, or scene evolution.

We close this gap by introducing TOSM [5]–[7], a structured world representation that tracks object-robot and object-object relationships semantically, enabling the system to reason not just where objects are, but what they are, how they relate, and which arm should handle them.

Second, VLA reasoning is computed once per chunk and never updated during the open-loop execution window. When a grasp destabilizes mid-chunk, the policy has no mechanism to detect it or recover principally.

We address this by building TOSM-VLA, a neurosymbolic bimanual manipulation framework that augments a probabilistic VLA policy with a deterministic, persistent semantic world model, enabling robust detection, diagnosis, and recovery from manipulation failures within the open-loop execution window.

II. RELATED WORKS

A. Imitation Learning for Robot Control

ALOHA teleoperation system and ACT algorithm [8] established the paradigm of low-cost bimanual teleoperation for imitation learning, demonstrating that learned policies could execute dexterous two-arm tasks from human demonstrations. ACT extended this with action chunking transformers for fine-grained bimanual control, while Diffusion Policy [9] introduced the action chunking paradigm that underpins modern visuomotor policies including our own. In all these systems, arm role assignment is determined by the human demonstrator during teleoperation and encoded implicitly into collected trajectories, leaving the policy with no understanding of why a particular arm was chosen. TOSM-VLA replaces this ad-hoc assignment with principled role resolution derived from object semantics prior to data collection.

B. Vision Language Model for Failure Reasoning

Recent work has shown that VLMs can be adapted for failure detection in robotic manipulation. AHA [10] frames failure detection as a free-form reasoning task, fine-tuning a VLM on procedurally generated failure trajectories to produce natural language explanations across diverse robots and tasks. I-FailSense [11] attaches lightweight classification heads to multiple internal layers of a post-trained VLM, achieving strong generalization to semantic misalignment failures. While both approaches demonstrate the viability of VLM-based failure reasoning, neither integrates into the real-time control loop. TOSM-VLA addresses this by embedding

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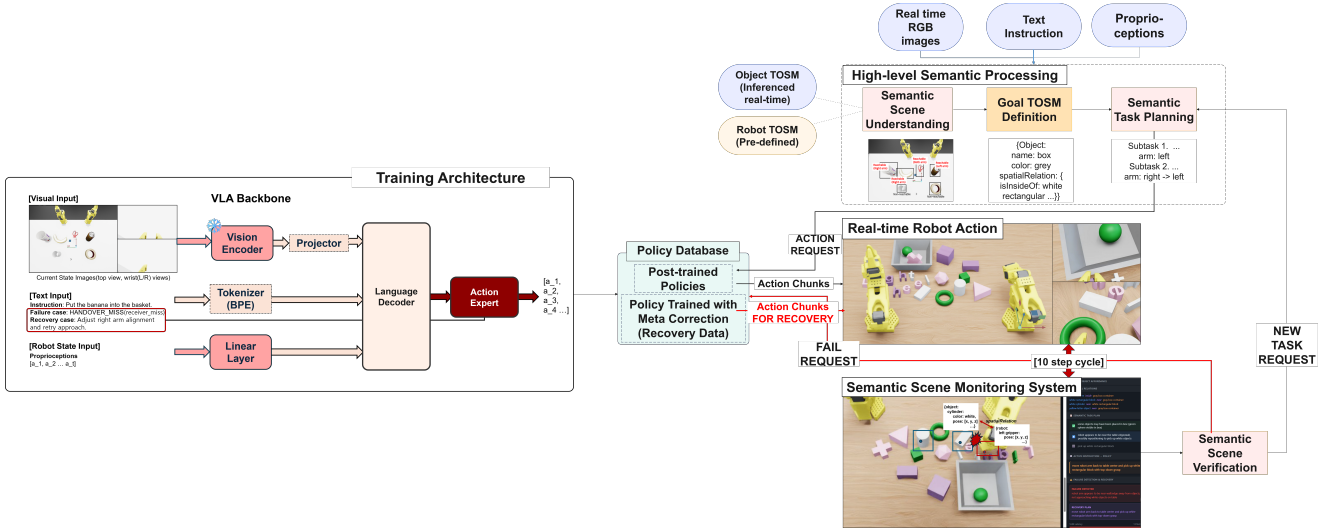


Fig. 1. Overall pipeline of TOSM-VLA, consists of *Training Architecture* including imitation learning method to obtain demonstration data for VLA policies, and *Inference Architecture* including High-level Semantic Processing and Semantic Scene Monitoring System governing Real-time Robot Action.

cause-conditioned failure detection and recovery directly within the async inference framework.

C. Vision Language Action Model for Bi-Manipulation

Bimanual manipulation poses unique challenges for VLA systems: coordinating two arms requires reasoning about multi-modal action distributions and inter-arm dependencies that single-arm policies cannot represent. RDT-1B [12] addresses data scarcity and action multi-modality with a 1.2B-parameter diffusion foundation model pre-trained across 46 multi-robot datasets, demonstrating zero-shot generalization on an ALOHA dual-arm platform. Bi-VLA [13] integrates vision, language, and action for bimanual dexterous manipulation, translating human instructions into coordinated two-arm executable code.

III. TOSM-VLA

As shown in Fig. 1, TOSM-VLA consists of two tightly coupled components: a training architecture that equips VLA policies with failure-aware reasoning, and an inference architecture that grounds execution in a persistent semantic world model.

Training. The VLA backbone receives three input streams: multi-view RGB images (top-view and bilateral wrist cameras), tokenized language instructions, and proprioceptive joint states processed through a vision encoder, BPE tokenizer, and linear projection respectively before being fused in the language decoder and action expert. The training corpus includes not only successful demonstrations but also failure-annotated trajectories with Meta Correction labels specifying responsible arm and grasp type for each recovery case, enabling the policy to learn cause-conditioned corrective behaviors directly from data. Meta Correction information is embedded in the text instruction input so that the policy can satisfy FAIL REQUEST conditions at inference time. Arm role assignment is encoded via the *ResponsibleArm* key, ensuring every demonstration carries

semantically grounded role annotations based on object properties and physical feasibility rather than arbitrary human choices—enabling consistent multimodal mapping across visual, linguistic, and proprioceptive streams.

Inference. At runtime, the system operates across three levels. RGB inputs from top-view and bilateral wrist cameras, together with text instructions, are processed by a high-level semantic processing module that constructs a real-time Object TOSM—encoding object identities, spatial relations, and goal states—against a pre-defined Robot TOSM, producing a structured subgoal queue for Semantic Task Planning. At the execution level, the policy database serves action chunks from the post-trained VLA policy selected based on the current task state. At the monitoring level, a Semantic Scene Monitoring System runs on a fixed 10-step cycle, continuously tracking spatial relations between objects and the gripper, and between objects themselves. Upon detecting a failure, a FAIL REQUEST is issued: the diagnosed failure case is passed to the policy database, which selects the Meta Correction policy to generate a recovery chunk ensuring the assigned subtask can be successfully completed. Successful task completion is verified through a Semantic Scene Verification module leveraging Vision Language Model before the next task request is accepted.

A. The Deterministic Model, TOSM

TOSM is a neuro-inspired, semantic modeling framework originally designed for cognitive robot navigation to emulate human-like cognitive abilities. The framework provides a pre-defined ontological schema that structures how robots represent, reason over, and act on environmental knowledge. As shown in Fig. 2, TOSM is organized into three layers: Symbol, Explicit, and Implicit.

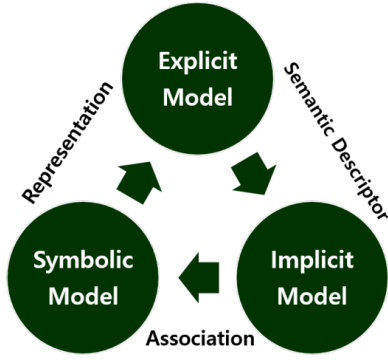


Fig. 2. Triplet Ontological Semantic Model (TOSM).

TOSM Properties for Dual-arm Robots. As listed in Table I, Robot’s symbolic properties establish the robot’s identity via name and ID. Explicit properties encode the robot’s physical constraints pre-defined from its URDF specification, including maximum payload, workspace reach, and gripper open width, which directly gate feasibility: an arm cannot be assigned to an object whose weight or position violates its limits. Implicit properties encode task-time inferred behavioral attributes. Grip type (top-down pinch, lateral pinch, and lateral power grasp) represents three manually defined grasp strategies derived from the robot’s joint range constraints, ensuring safe and feasible operation within the SO-101’s kinematic limits. The appropriate grip type is selected at task planning time by reasoning over the target object’s geometry and material encoded in the Object TOSM. At inference time, the Robot TOSM provides the source for two queries before any subgoal is committed: feasibility, asking whether this arm can reach and lift this object, and grasp selection, asking which grip type this object’s geometry requires. By resolving both at the representational level, TOSM-VLA constructs task plans grounded in physical reality and can reject or reassign subtasks when constraints are violated, without requiring additional training data.

TOSM Properties for Detected Objects. As listed in Table II, The Object TOSM follows the same three-layer schema as the Robot TOSM, with the schema structure and property types pre-defined at initialization. The Object TOSM is populated in real time from RGB observations via the Semantic Scene Understanding module, filling each TOSM property field as objects are detected and interpreted in the scene. Explicit properties encode perceptually grounded physical attributes including material, color, and shape, inferred directly from visual observation. These properties inform grasp feasibility reasoning: a steel cylinder affords different grip strategies than a soft deformable object of identical geometry. Implicit properties constitute the semantically rich layer of the Object TOSM. Boolean affordance flags such as *isMovable*, *canBePickedUp*, and *canBePushed* are derived from the combination of explicit physical attributes and robot capability constraints, determining which manipulation primitives are applicable to each object. *graspType* is resolved by jointly reasoning over object

shape and material against the arm’s explicit constraints encoded in the Robot TOSM. Critically, *spatialRelation* encodes the geometric relationships between objects as a dictionary map using predicates such as *isInsideOf* and *isRightTo*, providing the relational substrate on which goal state verification, subgoal sequencing, and failure diagnosis are performed. Together, the Robot and Object TOSMs form a jointly queryable world model that supports the full reasoning stack of TOSM-VLA: from task planning and arm assignment through real-time failure detection and cause-conditioned recovery.

TABLE I
MANIPULATOR’S TOSM REPRESENTATION

TOSM Datatype Properties		Datatype	Example
<i>Symbol</i>	<i>Name</i>	String	“so_arm_101_follower”
	<i>ID</i>	Int	1
<i>Explicit</i>	<i>Max_payload(g)</i>	Int	500
	<i>Reach(mm)</i>	Int	400
	<i>Gripper_Open_Width(mm)</i>	Int	80
<i>Implicit</i>	<i>Grip_type</i>	String	Top_down_pinch_grasp
			Lateral_pinch_grasp
			Lateral_power_grasp

TABLE II
OBJECT’S TOSM REPRESENTATION

TOSM Datatype Properties		Datatype	Example
<i>Symbol</i>	<i>Name</i>	String	“master_chef_can”
	<i>ID</i>	Int	0
<i>Explicit</i>	<i>Material</i>	String	“steel”
	<i>Color</i>	String	“blue”
	<i>Shape</i>	String	“cylinder”
<i>Implicit</i>	<i>isMovable</i>	Boolean	true
	<i>canBePickedUp</i>	Boolean	True
	<i>ResponsibleArm</i>	String	“left”
	<i>canBePushed</i>	Boolean	True
	<i>graspType</i>	String	Lateral_power_grasp
	<i>spatialRelation</i>	Dict., map	“isInsideOf”: box “isRightTo”: cup

B. High-level Semantic Processing

Semantic Scene Understanding serves as the perceptual entry point of TOSM-VLA, constructing and maintaining the Object TOSM in real time from multi-view RGB observations. As illustrated in Fig. 3, the information of each detected object is mapped to its TOSM representation. For example, the white cup is identified as a plastic cylinder with *canBePickedUp*: true, *ResponsibleArm*: right, and *graspType*: top_down_pinch_grasp, while the master chef can resolves to *ResponsibleArm*: left and *graspType*: lateral_power_grasp given its steel material and cylindrical geometry.

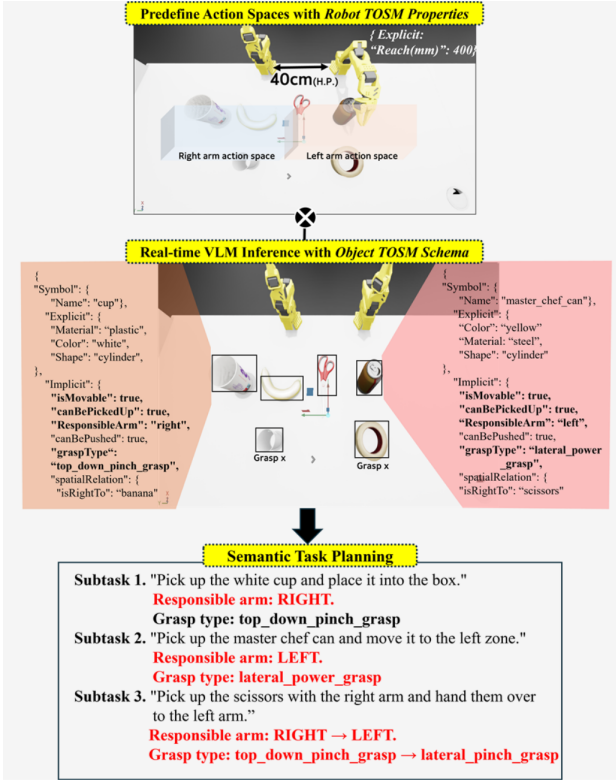


Fig. 3. Semantic Scene Understanding and Task Planning pipeline.

Affordance flags and grasp type are resolved relationally against the pre-defined Robot TOSM, cross-referencing each object’s explicit physical attributes against each arm’s payload, reach, and gripper width constraints, rather than assigned as fixed labels. The resulting Object TOSM is consumed by the Semantic Task Planning module, which decomposes the language instruction into a structured subgoal queue. A Goal TOSM is first defined by identifying the target *spatialRelation* from the instruction—e.g., *isInsideOf*: grey box—and cross-referencing it against the current scene state. The subgoal queue is then populated by sequencing the manipulation primitives required to transition from the current TOSM state to the goal state. As shown in Fig. 3, this produces subtasks with fully resolved arm assignments and grasp strategies, each verified by the Semantic Scene Monitoring System before the next subgoal is committed.

C. Recovery from Failure

Despite accurate task planning, execution failures are inevitable in real-world bimanual manipulation. As illustrated in Fig. 4, we identify four principal failure cases diagnosed through distinct TOSM relational signatures: Grasping Wrong Object (gripper contact mismatches the subgoal target), Wrong Orientation (gripper aperture insufficient for object geometry), Grasping Error (*spatialRelation* overlap decreases significantly without a place primitive having been executed), and Cooperation Error (gripper-object coupling persists beyond the expected handover point). As shown in Fig. 5, upon any failure the Semantic Monitoring System

issues a FAIL REQUEST while the robot client still holds remaining actions in the current chunk. The policy server cancels ongoing inference, re-plans with a recovery instruction derived from the current TOSM state, and serves a recovery chunk—ensuring the robot transitions directly to corrective execution with only inference latency as interruption, without requiring task restart.

IV. EXPERIMENTS

A. Experimental Setup

We evaluate TOSM-VLA on a dual the RobotStudio’s SO-101 bimanual platform across a desk organization task. The long horizon task instruction requires the system to perform sequential pick-and-place, and inter-arm handover. All experiments are conducted over 20 trials per condition.

B. Results

Table III reports task completion results. Semantic Task Planning achieved 90% success; Subtask 1 reached 80% (16/20) and Subtask 2 65% (13/20), reflecting the higher coordination complexity of handover, yielding an overall completion rate of 50%. Table IV reports failure detection and recovery performance: 14 of 17 observed failures were correctly detected (82%), with 3 missed detections attributable to brief slip events falling below the 10-step monitoring threshold. Of detected failures, the Meta Correction policy successfully recovered in 9 cases (64%), with 3 false positives triggering unnecessary queue flushes due to gripper deceleration being misclassified as contact loss. As illustrated in Fig. 6, the Semantic Monitoring System visualizes real-time *spatialRelation* between the gripper and target objects, displaying the active instruction and recovery plan alongside the TOSM-based scene state to support transparent failure diagnosis during execution.

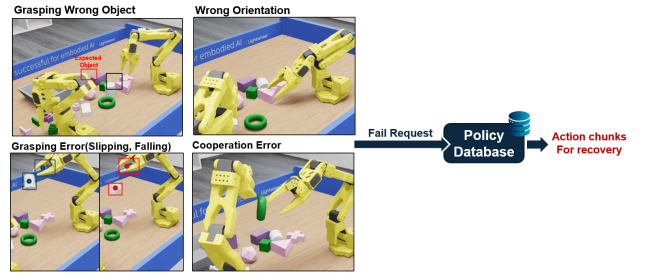


Fig. 4. Recovery from Failure Module.

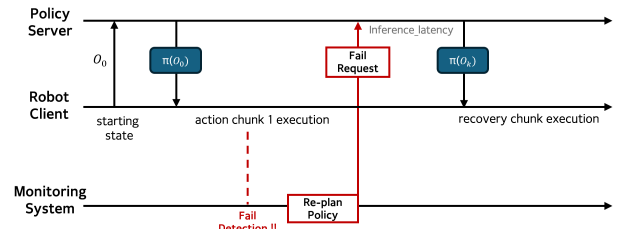


Fig. 5. Asynchronous Inference.

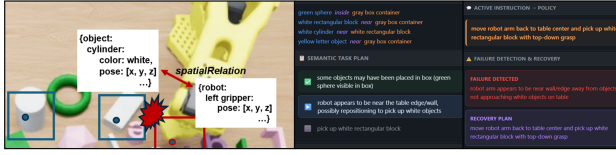


Fig. 6. TOSM-based Semantic Monitoring System.

TABLE III
PRELIMINARY RESULT—SORTING TASK ($n = 20$)

Components	Success Rate
Semantic Task Planning	18/20 (90%)
Subtask 1: Pick & Place	16/20 (80%)
Subtask 2: Handover and Place	13/20 (65%)
Full Task (Long-horizon)	10/20 (50%)

TABLE IV
FAILURE DETECTION & RECOVERY ($N = 20$)

Components	Success Rate
Failure Detection Accuracy	14/17 (82%)
Recovery Success Rate	9/14 (64%)
False Positive Rate	3/20 (15%)

V. CONCLUSIONS

We presented TOSM-VLA, a neurosymbolic framework coupling a persistent TOSM-based world model with a VLA policy, evaluated on a dual SO-101 platform across a long-horizon sorting task with 90% semantic task planning accuracy, 82% failure detection rate, and 50% full-task completion. Future work will integrate tactile and force-aware sensing into the monitoring loop to enable earlier failure detection within each of the four defined failure modes, before relational violations become visible in the camera feed. We will also extend the pre-defined ontological schema to new manipulation domains, increase trial scale, and validate across more complex cooperative bimanual scenarios.

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